OR-AS Operations Research Applications and Solutions	Case Name: **Social Apartments Ypres (2)**	Sector	**Construction (residential building)**
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	**Matilde Casado and Margarida Antunes**		User defined distributions
Date	**2015**	Project Control	Automatic tracking
File Name	**C2015-31 Social Apartments Ypres (2)**		Tracking based on user input

1. Project description

Project authenticity

The project involves demolition, foundation work, masonry, roofing, utilities installation, interior finishes, and landscaping.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	29
Planned Duration (PD)	271 days*
Budget At Completion (BAC)	1 310 723.5€
Renewable Resources	-
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	32%
Activity Distribution (AD)	49%
Length of Arcs (LA)	23%
Topological Float (TF)	43%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	16.0	15.5	-2.2
CRI-rho	15.5	15.1	-2.1
CRI-tau	11.2	11.0	-2.2

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	39.1	42.1	-0.4
SI	55.0	41.7	0.2
SSI	10.6	19.5	-3.2
CRI-r	14.1	16.9	-3.6
CRI-rho	12.9	17.1	-3.8
CRI-tau	9.8	14.2	-3.8

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	10.1	7.1
PV - SPI	27.5	26.2
PV - SCI	29.1	25.7
ED - 1	13.1	11.5
ED - SPI	27.1	25.9
ED - SCI	27.1	25.4
ES - 1	9.6	7.5
ES - SPI(t)	23.0	22.4
ES - SCI(t)	23.2	22.3

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	3.4	0.0
CPI	4.5	0.2
SPI	18.9	18.0
SPI(t)	17.3	16.8
SCI	19.2	17.6
SCI(t)	17.8	16.5
0.8 CPI + 0.2 SPI	7.7	6.6
0.8 CPI + 0.2 SPI(t)	7.1	5.9

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Automatic tracking was performed over 21 tracking periods with a length of 1 month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on simulation results.

Authenticity assessment is not relevant here as it is not possible to introduce any kind of tracking information obtained from the actual project owner when performing automatic tracking. Activity durations and corresponding costs were generated based on the distribution profiles described in section “2.2. Risk Analysis”.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	1868.5	-236636.6	-415.0	1.0	0.9	0.9	0.9
std dev	9331.2	263176.8	403.2	0.1	0.6	0.4	0.1
final	28537.5	0.0	-784.0	1.0	1.0	0.7	1.0

2.3.3.2. Time forecasting

PD	271 days	Real Duration	369 days	Late	36.2%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	319.3	4.4	1.4	1.4
PV - SPI	320.0	4.4	1.4	1.4
PV - SCI	320.7	4.4	1.3	1.3
ED - 1	321.4	4.5	1.3	1.3
ED - SPI	322.1	4.6	1.3	1.3
ED - SCI	322.9	4.6	1.3	1.3
ES - 1	323.6	4.5	1.3	1.3
ES - SPI(t)	324.3	4.4	1.3	1.3
ES - SCI(t)	325.0	4.4	1.3	1.3

2.3.3.3. Cost forecasting

BAC	1 310 723.5€	Real Cost	1,282,186 €	Under Budget	-2.2%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	1308855.0	9331.2	0.1	-0.1
CPI	1328688.8	118123.7	0.2	-0.2
SPI	1543765.4	501105.5	1.0	-1.0
SPI(t)	1413815.3	285910.6	0.5	-0.5
SCI	154627.4	482483.2	1.0	-1.0
SCI(t)	1420881.9	258664.1	0.5	-0.5
0.8 CPI + 0.2 SPI	1322145.2	72021.2	0.1	-0.1
0.8 CPI + 0.2 SPI(t)	1321459.5	49032.1	0.1	-0.1